

1. Install FileZilla Server on your Windows machine, create a user account, and set up a shared folder for FTP access.

Install and Configure FileZilla Server on Windows

1. Download FileZilla Server

1. Open a web browser.
2. Download **FileZilla Server** from the official website:
[FileZilla Server official download](#)
3. Run the downloaded installer.

2. Install FileZilla Server

1. Accept the license agreement.
2. Select the default installation components.
3. Keep the default installation folder unless you have a reason to change it.
4. For the server startup option, select **Start as a service** so the FTP server can run automatically.
5. Complete the installation.

3. Open FileZilla Server

1. Launch **FileZilla Server** or **FileZilla Server Interface**.
2. Connect to the local server using the administration interface.
3. Use the administrator password configured during installation.

4. Create an FTP User

1. Open **Users** in the FileZilla Server interface.
2. Click **Add**.
3. Enter a username, for example:

ftpuser

4. Enable **Password** authentication.
5. Set a strong password.
6. Save/apply the settings.

5. Create a Shared Folder

First create a folder in Windows, for example:

C:\FTPShare

Then in FileZilla Server:

1. Select the ftpuser account.
2. Open the **Mount points / Shared folders** section.
3. Add C:\FTPShare as the user's accessible folder.
4. Set it as the user's **home directory**.
5. Give the user appropriate permissions:
 - **Read** – download files.
 - **Write** – upload files.
 - **Delete** – delete files.
 - **Create** – create folders/files.

For a basic test, **Read + Write** is sufficient.

6. Allow FileZilla Through Windows Firewall

If another computer needs to connect to the FTP server:

1. Open **Windows Defender Firewall**.
2. Select **Advanced settings**.
3. Create an inbound rule allowing the FTP service/port configured by FileZilla.
4. The traditional FTP control port is **TCP 21**.
5. Apply the rule to the appropriate network profile.

For passive FTP, also configure a passive port range in FileZilla Server and allow that TCP range through the firewall.

7. Test the FTP Server

From the same computer, you can test using FileZilla Client:

- **Host:** 127.0.0.1
- **Username:** ftpuser
- **Password:** the password you created
- **Port:** 21 (if using the default FTP port)

2. Use FileZilla Client to connect to your FTP server using the credentials you created, and upload any image file from your computer to the shared folder.

Upload an Image Using FileZilla Client

1. Install FileZilla Client

Download and install **FileZilla Client** from the official website:

[FileZilla Client download](#)

2. Connect to Your FTP Server

Open FileZilla Client and enter:

Field	Value
Host	127.0.0.1
Username	ftpuser
Password	Your created FTP password
Port	21

Click **Quickconnect**.

If the FTP server is on another computer, replace 127.0.0.1 with that computer's IP address, for example 192.168.1.10.

3. Select an Image

In the **Local site** section on the left:

1. Browse to the location of an image on your computer.
2. For example:
C:\Users\YourName\Pictures
3. Select an image such as:
music.jpg

4. Upload the Image

On the **Remote site** side, open your FTP shared folder, for example:

Then:

1. Right-click music.jpg.

2. Select **Upload**.
3. Wait for the transfer to finish.

You should see the file under **Successful transfers** at the bottom of FileZilla.

5. Verify the Upload

Open your Windows shared folder:

C:\FTPShare

3. Download the uploaded file from your FTP server to a different folder on your computer using FileZilla Client, and verify the file size matches the original.

Download the Uploaded File Using FileZilla Client

1. Open FileZilla Client

Open **FileZilla Client** and connect to your FTP server using the credentials created earlier:

Field	Value
Host	127.0.0.1
Username	ftpuser
Password	Your FTP password
Port	21

Click **Quickconnect**.

2. Locate the Uploaded File

On the **Remote site** side (right side), open the shared FTP folder.

For example, locate:

music.jpg

3. Create a Different Download Folder

Create a new folder somewhere else on your computer, for example:

C:\FTPDownloads

In FileZilla's **Local site** section (left side), navigate to:

C:\FTPDownloads

4. Download the File

1. On the **Remote site**, right-click music.jpg.
2. Select **Download**.
3. FileZilla will transfer the file to:

C:\FTPDDownloads

4. Check the **Successful transfers** section to confirm the download completed.

5. Verify the File Size

Compare the original image with the downloaded image.

Original:

C:\FTPShare\music.jpg

Downloaded:

C:\FTPDDownloads\music.jpg

Right-click each file → **Properties** → compare **Size**.

Example:

File	Location	Size
Original	C:\FTPShare\music.jpg	2.45 MB
Downloaded	C:\FTPDDownloads\music.jpg	2.45 MB

If both sizes are identical, the file was downloaded successfully without a size difference.

4. Research and list 3 real-world scenarios where SNMP is used in network management for apps or services you use (such as ISPs, campus WiFi, or large websites).

Scenario	How SNMP is used	Example
1. ISP / Internet Service Provider	ISPs monitor large numbers of routers and switches using SNMP. They can check interface traffic, errors, device availability, and hardware health. If a link or device fails, the monitoring system can generate an alert.	When your home Internet becomes slow or an ISP router/link fails, network administrators can use SNMP monitoring to identify overloaded interfaces or failed equipment. (Netdata)
2. College / Campus Wi-Fi	A university can monitor hundreds of Wi-Fi access points through a central network-management system. SNMP can provide information about AP availability and network conditions, while SNMP traps can report important events.	If a Wi-Fi access point in a classroom goes offline, administrators can receive an alert and troubleshoot it without physically checking every AP. Cisco and WatchGuard APs support SNMP-based monitoring. (WatchGuard Technologies)
3. Large websites / Data centers	Websites and online services depend on large numbers of switches, routers, firewalls, and load-balancing/network devices. SNMP monitoring helps operations teams track network-device health and traffic and detect failures before they cause larger outages.	A data-center monitoring system can detect a heavily utilized switch interface or a failed network device and alert the network team. Tools such as Zabbix can monitor SNMP-enabled routers and switches. (Zabbix)

5. Create a table comparing FTP and HTTP across at least 5 points: port number, security, data transfer method, use cases, and authentication.

Point	FTP (File Transfer Protocol)	HTTP (HyperText Transfer Protocol)
1. Port Number	TCP 21 for control; TCP 20 commonly used for active-mode data	TCP 80
2. Security	Basic FTP is not encrypted . Secure alternatives include FTPS and SFTP	Basic HTTP is not encrypted . HTTPS uses TLS encryption and normally uses TCP 443
3. Data Transfer Method	Designed specifically for uploading and downloading files between a client and server	Uses request-response communication between a client and web server
4. Use Cases	File sharing, website file management, transferring files to/from servers, backups	Websites, web applications, APIs, images, videos, and other web content
5. Authentication	Supports username and password authentication; can also allow anonymous access	Authentication is usually handled by the web application using methods such as passwords, tokens, cookies, or HTTP authentication
6. Connection	Traditionally uses a separate control connection and data connection	Typically uses one connection for HTTP requests/responses, with modern HTTP versions optimizing connection reuse
7. Example	Uploading assignment.pdf to a college FTP server using FileZilla	Opening a website such as a college portal in Chrome